

Mahmoud Moftah Mohamed

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PROFESSIONAL SUMMARY

Mechatronics Engineering student focused on embedded systems, autonomous flight, and control engineering. Hands-on experience in STM32 firmware development, dynamic modeling and controller tuning in MATLAB/Simulink, and UAV guidance and system integration through TEKNOFEST projects and real flight testing. Additional experience in ROS2/Gazebo simulation and industrial automation, with a career focus on aerospace and defense applications.

WORK EXPERIENCE

ABRA Student Team – TEKNOFEST UAV Projects

Oct 2025 – Present

Embedded Systems Engineer

Sakarya, Turkey

- Developed STM32-based firmware using GPIO, timers/PWM, and UART/SPI/I2C/CAN interfaces; performed hardware debugging and interface verification using ST-Link and a logic analyzer.
- Served as **Guidance & Control Lead** for the **TEKNOFEST 2026 Fighter UAV Hunter Drone Competition**.
- Served as **Autonomous Flight Software Engineer** for the **TEKNOFEST 2026 International Unmanned Aerial Vehicle Competition**.

YTG SANCAKTAR Student Team – TEKNOFEST Unmanned Surface Vehicle Competition

Oct 2024 – Jun 2025

Simulation Engineer | ROS2 / Gazebo

Sakarya, Turkey

- Developed a ROS2/Gazebo simulation environment for an unmanned surface vehicle, integrating URDF/XACRO models, TF frames, and simulated sensors to support SLAM, localization, and autonomous navigation testing.
- Member of the team that received the **“Best Software Team”** award at TEKNOFEST.

TECHNICAL PROJECTS

ABRA Hunter – TEKNOFEST 2026 Fighter UAV Hunter Drone Competition

Feb 2026 – Sep 2026

Guidance & Control Lead | Python / MATLAB / Betaflight / CRSF

FPV quadrotor designed to approach, visually track, and autonomously intercept a moving fixed-wing UAV using server telemetry and onboard camera imagery.

- Developed a **50 Hz mission and control architecture** covering takeoff, tracking, engagement, and landing; implemented **Pure Pursuit and Proportional Navigation** to generate attitude and altitude references for Betaflight ANGLE mode.
- Identified the drone's vertical dynamics from flight logs and developed and tuned the high-level altitude-control architecture, including altitude and vertical-velocity control with hover feed-forward, tilt compensation, anti-windup, and slew-rate limiting.
- Integrated team-developed **UKF/IMM-based state estimation and YOLO tracking modules** with telemetry, competition-server communication, ground-station controls, and manual RC handover; tested the integrated system through autonomous flight trials and flight-log analysis.

YUTPA UAV – TEKNOFEST 2026 International Unmanned Aerial Vehicle Competition (Rotary Wing)

Autonomous Flight Software Engineer | Python / MAVLink / ArduPilot / Gazebo

Feb 2026 – Sep 2026

- Developed MAVLink-based communication using pymavlink between a Raspberry Pi 5 and SpeedyBee F405 V4 flight controller via USB, implementing flight-mode transitions, arming/takeoff sequencing, command validation, and mission control.
- Implemented an **∞ -flight algorithm** generating waypoints dynamically from two GPS pole coordinates, including waypoint generation and upload, AUTO-mode mission execution, and landing using MAV_CMD_NAV_LAND.
- Developed autonomous target-alignment logic, GUIDED-mode transitions, and servo-triggered payload release; tested the mission workflow in **ArduPilot SITL and Gazebo Harmonic** with a GStreamer UDP camera stream.

STM32-Based Robotic Arm – Cascaded PID Position Control2025

Embedded Software Lead | C / STM32F446RE / MATLAB / Simulink

- Developed STM32F446RE firmware for a cascaded position-control architecture with a 100 Hz outer position loop and 1 kHz inner velocity loop.
- Integrated quadrature-encoder feedback, PWM motor control through an L298N driver, timer-based control execution, and UART communication for debugging and monitoring.
- Supported controller design and pre-hardware validation using MATLAB/Simulink, with the system targeting $\pm 2^\circ$ positioning accuracy.

Industrial Automation – Conveyor Sorting & Palletizing System2025

PLC / HMI Developer | Siemens S7-1214C / TIA Portal / Factory IO

- Developed and simulated a conveyor sorting and palletizing system using **TIA Portal, PLCSIM, and Factory IO**, implementing weight-based classification and automated routing.
- Implemented Ladder Logic for conveyor sequencing, speed control, item counting, and a four-step pick-and-place sequence.
- Designed a **KTP700 Basic HMI** with operator controls, live weight and speed monitoring, production counters, and equipment status indicators.

EDUCATION

Sakarya University of Applied Sciences (SUBÜ)Sakarya, Turkey

B.Sc. in Mechatronics EngineeringExpected Graduation: Jul 2027

Academic Focus & Relevant Coursework: Embedded Systems, Microcontrollers, Digital Systems, Control Systems, Signals and Systems, Robotics, Electronics, and Simulation.

SKILLS

- Embedded Systems & Firmware:** STM32 (bare-metal & HAL), Embedded C, Arduino, Raspberry Pi, UART/SPI/I2C/CAN, PWM, interrupt-driven programming, FreeRTOS (basic), ST-Link, logic analyzer, Proteus
- Control & Modeling:** MATLAB/Simulink, system identification, dynamic modeling, PID and cascaded control, controller tuning, anti-windup, signal filtering
- UAV Autonomy & Guidance:** Pure Pursuit, Proportional Navigation, mission state machines, ArduPilot, Betaflight, MAVLink/pymavlink, state-estimation and vision integration, SITL, Mission Planner
- Programming & Tools:** Python, C++, OOP, Git/GitHub, Linux
- Simulation & Robotics:** ROS/ROS2, Gazebo Harmonic, RViz, URDF/XACRO, SLAM (basic)
- Industrial Automation:** Siemens S7-1200, TIA Portal, Ladder Logic, Factory IO, HMI development (KTP700 Basic)
- Mechanical Design & Analysis:** CATIA, AutoCAD, Abaqus (FEA)

LANGUAGES

Arabic: Native | Turkish: C2 | English: C1

EXTRACURRICULAR

Bassma Team – Student Community InitiativeOct 2024 – Present

Co-Founder & Operations Coordinator (Volunteer)Sakarya, Turkey

- Organized educational, cultural, and volunteer programs; coordinated event planning, logistics, team responsibilities, and community communications.